INTRODUCTION

R.D. Engineering College, Ghaziabad

Topic - Bidirectional Visitor Counter Using Arduino UNO

Mini project

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Content

AIM of Project

> The aim of our project is to make a controller which can sense if any person enters the it lights up the room and automatically and also counts how many person are entering the room or going out of it.

> It is made to prevent unwanted electric power waste in schools, colleges, office and houses . This whole process is operated automatically by its sensors .

Introduction

> In today’s world, there is a continuous need for automatic appliances .

> The objective of this project is to make a controller based model to count number of persons visiting particular room and accordingly light and display it on LED.

> Here we are using sensor and controller to count the no of person in the room and display it on LCD.

> If there is at least 1 person in the room, the LED will glow else it will remain off.

Components Required

> Hardware components

> Arduino UNO

> Resistance

> IR Sensor module

> 16*2 LCD display

> Bread Board

> connecting wire

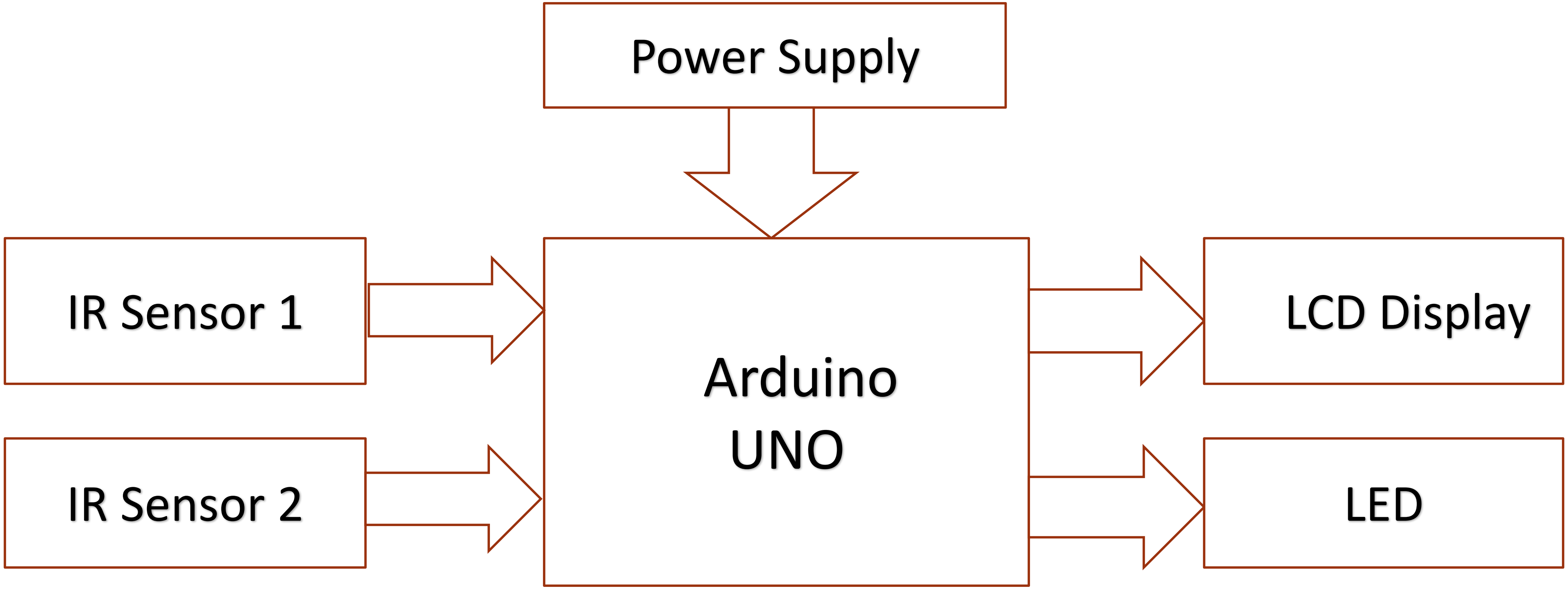
> LED

> BC547 Transistor

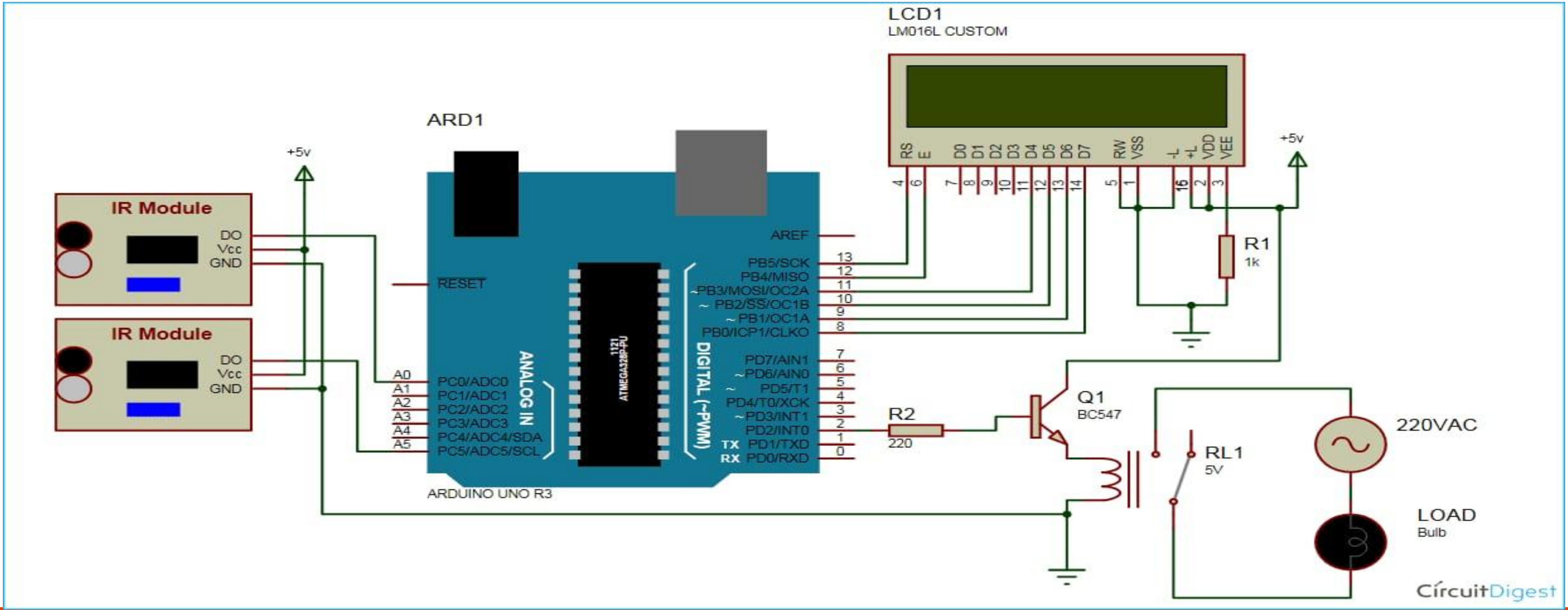
>Bulb Holder

>LED Bulb

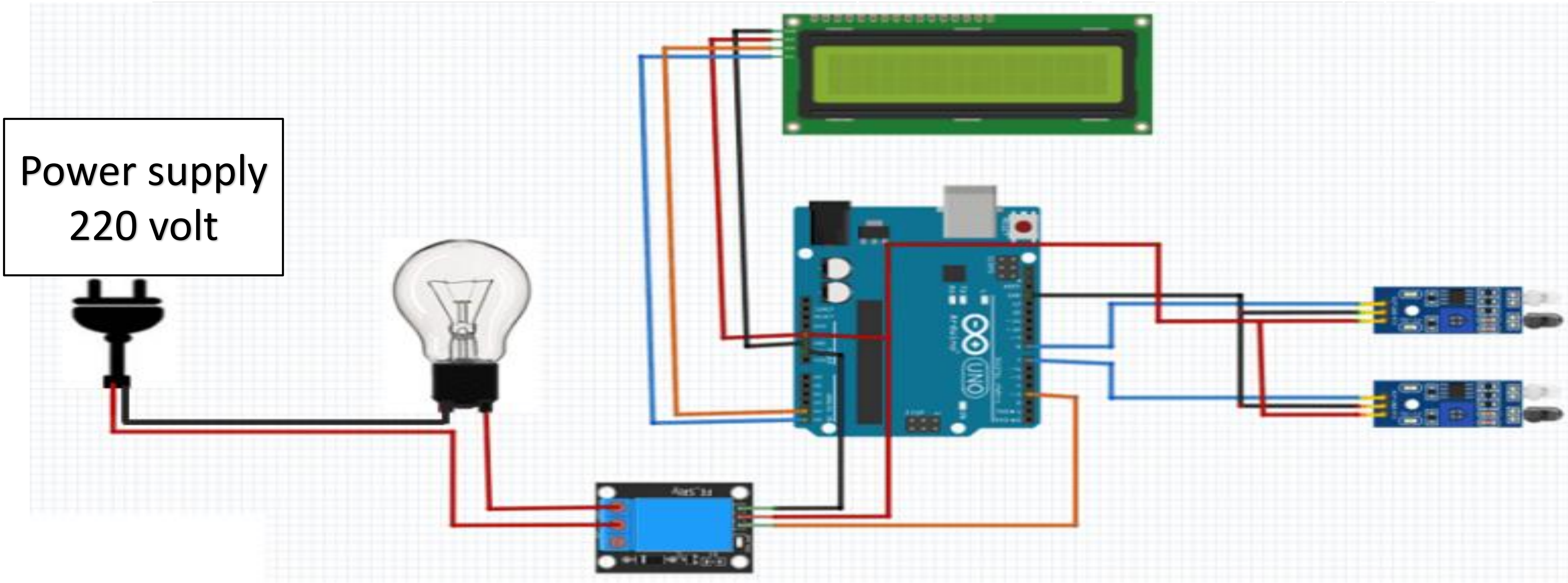
Block Diagram



Schematic Diagram



Implement of Schematic Diagram



Working

>The IR Sensor continuously senses the presence of any obstacles (a person in our case).

> If sensor 1 senses a person, it performs the controller that a person has entered so that controller can increment the count.

> At the same time it gives a delay of 1 sec so that the person can cross the sensor 2 and the count is maintained correctly

Working

> when a person exits, the sensor 2 informs the controller to decrement the count, similarly it also provides a delay of 1 sec to maintain count properly.

> The count is displayed on LCD by the controller.

> if there is at least 1 person is inside the hall, an LCD will glow otherwise it is off.

Advantage

> Can be used for automatic room light control.

> It will help to save electricity . When no one is there in room the appliances will be off.

> In school/companies it will help to check if somebody is there in the zone or not. If the data on display unit is zero the security guaeds can shut the gate easily.

Limitation

> It is used only when one person cuts the rays of the sensor hence cannot be used when to or more persons cross the door simultaneously.

> When anybody is inside the room and we need to switch off the power then we have to do it manually. So , in this case we fail to automatically control the light.

Application

> This circuit can be used domestically to get an indication of number of person entering a party.

> It can be used at home and other places to keep a check on the number of person entering a secured place.

> It can also be used as home automation system to ensure energy saving by switching on the loads and fans only when needed.


THANK
YOU!